

- The optimum descent rate is dependent upon ground speed. A rule of thumb is to multiply half of ground speed by 10. For example, a 130-knot ground speed should result in a (65 times 10) 650 feet per minute descent rate. Typical descent rates fall between 500 and 700 feet per minute. An excessive vertical speed may indicate a problem with the approach.

Every approach should be evaluated at 500 feet. In a typical jet airplane, this is approximately 1 minute from touchdown. If the approach is not stabilized at that height, a go-around should be initiated. [Figure 16-18]

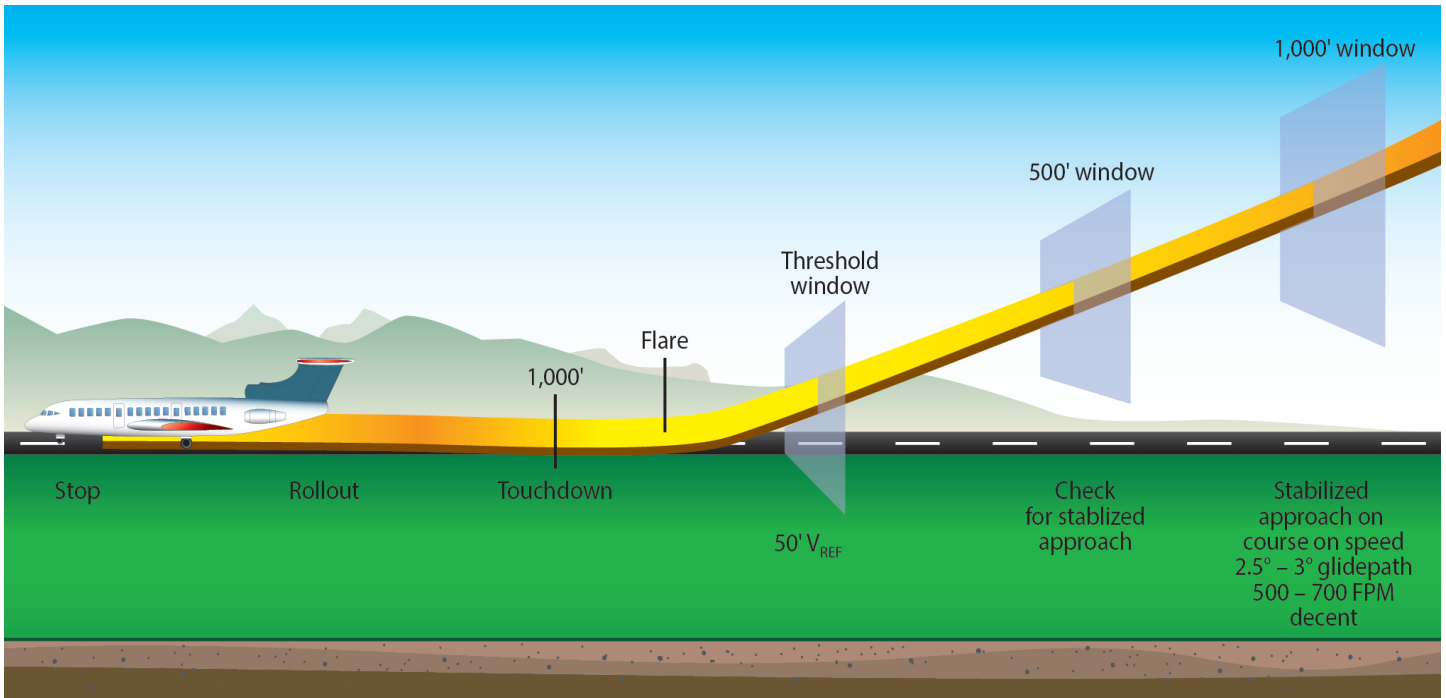


Figure 16-18. Stabilized approach.

Approach Speed

Any speed deviation on final approach should be detected immediately and corrected. With experience, the pilot is able to detect the onset of an increasing or decreasing airspeed trend, which normally can be corrected with a small adjustment. It is imperative the pilot does not allow the airspeed to decrease below V_{REF} or a high sink rate can develop. If an increasing sink rate is detected, it should be countered by increasing the AOA and simultaneously increasing thrust to counter the extra drag. The degree of correction depends on how much the sink rate needs to be reduced. For small amounts, smooth and gentle, almost anticipatory corrections are sufficient. For large sink rates, drastic corrective measures would be required that, even if successful, would destabilize the approach.

A common error in the performance of approaches in jet airplanes is excess approach speed. Excess approach speed carried through the threshold window and onto the runway increases the minimum stopping distance required by 20–30 feet per knot for a dry runway and 40–50 feet for a wet runway. Worse yet, the excess speed increases the chances of an extended flare, which increases the distance to touchdown by approximately 250 feet for each excess knot in speed.

Proper speed control on final approach is of primary importance. The pilot should anticipate the need for speed adjustment so that only small adjustments are required, and the airplane arrives at the approach threshold window exactly on speed.

Glidepath Control

The optimum glidepath angle is about 3°. On visual approaches, pilots may have a tendency to make flat approaches. A flat approach, however, increases landing distance and should be avoided. For example, an approach angle of 2° instead of a recommended 3° adds 500 feet to landing distance.

A more common error is excessive height over the threshold. This could be the result of an unstable approach or a stable but high approach. It also may occur during a nonprecision instrument approach where the missed approach point is close to or at the runway threshold. Regardless of the cause, excessive height over the threshold most likely results in a touchdown beyond the normal aiming point. An extra 50 feet of height over the threshold adds approximately 1,000 feet to the landing distance. The airplane should arrive at the approach threshold window exactly on altitude (50 feet above the runway).

The Flare

The flare reduces the approach rate of descent to a more acceptable rate for touchdown. Unlike light airplanes, a jet airplane should be flown onto the runway rather than “held off” the surface as speed dissipates. A jet airplane is aerodynamically clean even in the landing configuration, and its engines still produce residual thrust at idle rpm. Holding it off during the flare in an attempt to make a smooth landing greatly increases landing distance. A firm landing is normal and desirable. A firm landing does not mean a hard landing, but rather a deliberate or positive landing.

For most airports, the airplane passes over the end of the runway with the landing gear 30–45 feet above the surface, depending on the landing flap setting and the location of the touchdown zone. It takes 5–7 seconds from the time the airplane passes the end of the runway until touchdown. The flare is initiated by increasing the pitch attitude just enough to reduce the sink rate to 100–200 fpm when the landing gear is approximately 15 feet above the runway surface. In most jet airplanes, this requires a pitch attitude increase of only 1° to 3° . The thrust is smoothly reduced to idle as the flare progresses.

The normal speed bleed off during the time between passing the end of the runway and touchdown is just a few knots. Most of the decrease occurs during the flare when thrust is reduced. If the flare is extended (held off) while an additional speed is bled off, hundreds or even thousands of feet of runway may be used up. [Figure 16-19] The extended flare also results in additional pitch attitude, which may lead to a tail strike. It is, therefore, essential to fly the airplane onto the runway at the target touchdown point, even if the speed is excessive. A deliberate touchdown should be planned and practiced on every flight. A positive touchdown helps prevent an extended flare.

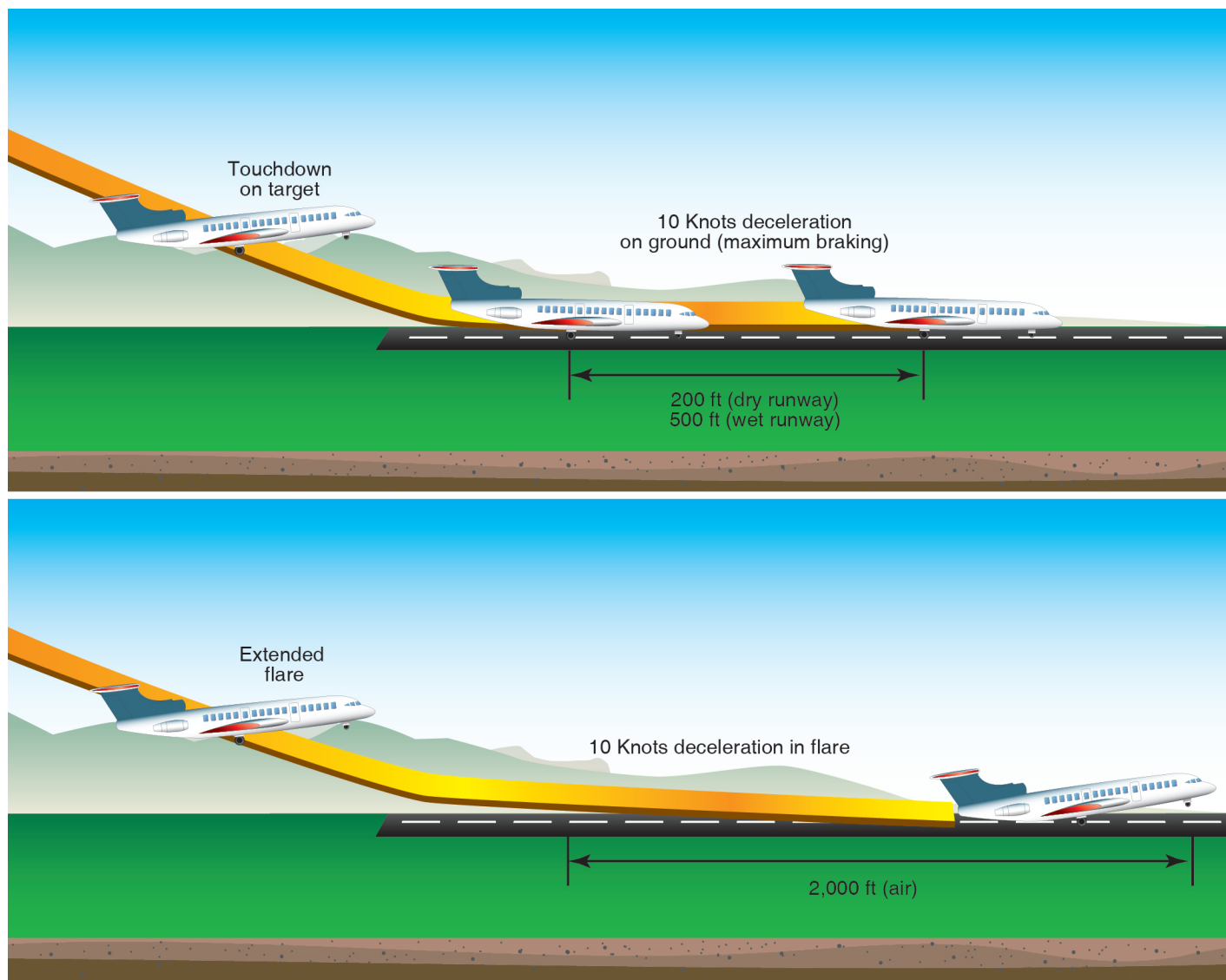


Figure 16-19. Extended flare.

Pilots should learn the flare characteristics of each model of airplane they fly. The visual reference cues observed from each airplane are different because window geometry and visibility are different. The geometric relationship between the pilot's eye and the landing gear is different for each make and model. It is essential that the flare maneuver be initiated at the proper height—not too high and not too low.

Beginning the flare too high or reducing the thrust too early may result in the airplane floating beyond the target touchdown point or may include a rapid pitch up as the pilot attempts to prevent a high sink rate touchdown. This can lead to a tail strike. The flare that is initiated too late may result in a hard touchdown.

Proper thrust management through the flare is also important. In many jet airplanes, the engines produce a noticeable effect on pitch trim when the thrust setting is changed. A rapid change in the thrust setting requires a quick elevator response. If the thrust levers are moved to idle too quickly during the flare, the pilot may need to make rapid changes in pitch control. If the thrust levers are moved more slowly, the elevator input can be more easily coordinated.

Touchdown and Rollout

A proper approach and flare positions the airplane to touch down in the touchdown target zone, which is usually about 1,000 feet beyond the runway threshold. Once the main wheels have contacted the runway, the pilot should maintain directional control and initiate the stopping process on the runway that remains in front of the airplane. The runway distance available to stop is longest if the touchdown was on target. The energy to be dissipated is least if there is no excess speed.

At the point of touchdown, the airplane represents a very large mass that is moving at a relatively high speed. The large total energy gets dissipated by the brakes, the aerodynamic drag, and the thrust reversers (if available). The nose-wheel should be lowered onto the ground immediately after touchdown because a jet airplane decelerates poorly when held in a nose-high attitude, and placing the nose-wheel tire(s) on the ground assists in maintaining directional control. Lowering the nose gear decreases the wing AOA, decreasing the lift, placing more load onto the tires, thereby increasing tire-to-ground friction. Landing distance charts for jet airplanes assume that the nose-wheel is lowered onto the runway within 4 seconds of touchdown.

There are only three forces available for stopping the airplane: wheel braking, reverse thrust, and aerodynamic braking. Of the three, the brakes are most effective and therefore the most important stopping force for most landings. When the runway is very slippery, reverse thrust and drag may be the dominant forces. Both reverse thrust and aerodynamic drag are most effective at high speeds. Neither is affected by runway surface conditions. Brakes, on the other hand, are most effective at low speed. The landing rollout distance depends on the touchdown speed, what forces are applied, and when they are applied. The pilot controls the what and when factors, but the maximum braking force may be limited by tire-to-ground friction.

The pilot should begin braking as soon after touchdown and wheel spin-up as possible, and smoothly continue the braking until stopped or a safe taxi speed is reached. However, caution should be used if the airplane is not equipped with a functioning anti-skid system. In such a case, heavy braking can cause the wheels to lock and the tires to skid.

Both directional control and braking utilize tire ground friction. They share the maximum friction force the tires can provide. Increasing either subtracts from the other. Understanding tire ground friction, how runway contamination affects it, and how to use the friction available to maximum advantage is important to a jet pilot.

Spoilers should be deployed immediately after touchdown because they are most effective at high speed. Timely deployment of spoilers increases drag significantly, but more importantly, they spoil much of the lift the wing is creating, thereby causing more of the weight of the airplane to be loaded onto the wheels. The spoilers increase wheel loading, which increases the tire ground friction force making the maximum tire braking forces available.

Like spoilers, thrust reversers are most effective at high speeds and should be deployed quickly after touchdown. However, the pilot should not command significant reverse thrust until the nose-wheel is on the ground. If the reversers deploy asymmetrically resulting in an uncontrollable yaw toward the side with more reverse thrust, the pilot needs whatever nose-wheel steering is available to maintain directional control. When runway length is not a factor, using idle reverse thrust may be adequate.

Jet Airplane Systems and Maintenance

All FAA-certificated jet airplanes are certificated under Title 14 of the Code of Federal Regulations (14 CFR) part 25, which contains the airworthiness standards for transport category airplanes. The FAA-certificated jet airplane is a highly sophisticated machine with proven levels of performance and guaranteed safety margins. The jet airplane's performance and safety margins can only be realized, however, if the airplane is operated in strict compliance with the procedures and limitations contained in the FAA-approved AFM for the particular airplane. Furthermore, in accordance with 14 CFR part 91, section 91.213(a), a turbine-powered airplane does not qualify to takeoff with inoperable instruments or equipment installed unless, among other requirements, an approved Minimum Equipment List (MEL) exists for that aircraft, and the aircraft is operated under all applicable conditions and limitations contained in the MEL (section 91.213(a)(5)).

Minimum Equipment List

The MEL serves as a reference guide for dispatchers and pilots to determine whether takeoff of an aircraft with inoperative instruments or equipment is authorized under the provisions of applicable regulatory requirements.

The operator models the MEL after the FAA's Master MEL (MMEL) for each type of aircraft and the Administrator approves the MEL before its implementation. The MEL includes a "General" section, comprised of definitions, general policies, as well as operational procedures for flight crews and maintenance personnel. Each aircraft component addressed in the MEL is listed in an alphabetical index for quick reference. A table of contents further divides the manual in different chapters, each numbered for its corresponding aircraft system designation (i.e., the electrical system, also designated as system number 24, would be found in chapter 24 of the MEL).

Pilots may defer repair of items on those aircraft systems and components allowed by the approved MEL. Per 14 CFR part 91, section 91.213(a)(3)(ii), an MEL must provide for the operation of the aircraft with the instruments and equipment in an inoperable condition. If particular items do not allow for safe operation, they do not appear on the MEL and takeoff is not authorized until the item is adequately repaired or replaced (section 91.213(a)). In cases where repairs may temporarily be deferred, operation or dispatch of an aircraft whose systems have been impaired is often subject to limitations or other conditional requirements explicitly stated in the MEL. Such conditional requirements may be of an operational nature, a mechanical nature, or both.

Mechanical conditions outlined in the MEL may require precautionary pre-flight checks, partial repairs prior to departure, or the isolation of selected elements of the deficient aircraft system (or related interacting systems), as well as the securing of other system components to avoid further degradation in flight. The MEL may contain either a step-by-step description of required partial maintenance actions or a list of numerical references to the Maintenance Procedures Manual (MPM) where each corrective procedure is explained in detail. Procedures performed to ensure the aircraft can be safely operated are categorized as either operations procedures or maintenance procedures. The MEL will denote which by indicating an "O" or an "M" as appropriate.

If operational and mechanical conditions can be met, an authorized person makes an entry in the aircraft MEL Deferral Record and issues a temporary placard. This authorizes the operation for a limited time before permanent repairs take place. The placard is affixed by maintenance personnel or the flight crew onto or next to the instrument or control mechanism to remind the flight crew of any limitations.

The MEL only applies while the aircraft sits on the ground awaiting departure or takeoff. It is essentially a dispatching reference tool used in support of all applicable Federal Aviation Regulations. If dispatchers are not required by the operator's certificate, flight crews still need to refer to the MEL before dispatching themselves to ensure that the flight is planned and conducted within the operating limits set forth in the MEL. Once the aircraft leaves the ground, any mechanical failures should be addressed using the appropriate checklists and approved AFM, not the MEL. Although a pilot may refer to the MEL for background information and documentation, actions in flight should be based strictly on instructions provided by the AFM (i.e., Abnormal or Emergency sections).

Configuration Deviation List

A Configuration Deviation List (CDL) is used in the same manner as an MEL but it differs in that it addresses missing external parts of the aircraft rather than failing internal systems and their constituent parts. They typically include elements, such as service doors, power receptacle doors, slat track doors, landing gear doors, APU ram air doors, flaps fairings, nose-wheel spray deflectors, position light lens covers, slat segment seals, static dischargers, etc.

Chapter Summary

Some of the differences when transitioning from props to jets include:

- Engine intake suction and exhaust create a ground hazard.
- There is no propeller-induced lift when power increases.
- Engine spool up time from low power settings is longer.
- Swept wing stalls begin at the tips.
- Higher speeds require smaller and smoother flight control inputs.
- Descents require more planning and optimally occur at idle power.
- When descending at speeds above L/D_{MAX} , increasing speed increases rate of descent and descent angle.

There are many considerations for a pilot when transitioning to turbojet-powered airplanes. In addition to the information found in this chapter and type specific information that will be found in an FAA-approved Airplane Flight Manual, a pilot can find basic aerodynamic information for swept wing jets, considerations for operating at high altitudes, and airplane upset causes and general recovery procedures in the Airplane Upset Recovery Training Aid, Supplement, pages 1-14, and all of Section 2 found at www.faa.gov/other_visit/aviation_industry/airline_operators/training/media/ap_upsetrecovery_book.pdf.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 17: Transition to Light Sport Airplanes (LSA)

Introduction

The light-sport aircraft (LSA) concept broadens the access of flight to more people. LSA have been defined as a simple-to-operate, easy-to-fly aircraft; however, “simple-to-operate” and “easy-to-fly” do not negate the need for proper and effective training. This chapter introduces the light-sport category of airplanes and places emphasis on transition to a light-sport airplane.

Even though light-sport airplane flight may appear simple to an experienced pilot, a transition to a light-sport airplane should include the same methodical training approach as transitioning into any other airplane. A pilot seeking a transition into light-sport airplane flying should follow a systematic, structured light-sport airplane training course under the guidance of a competent instructor with recent experience in the specific training airplane.

Light-Sport Aircraft Background

Several groups were instrumental in the development and success of the LSA concept. These included the Federal Aviation Administration (FAA), Light Aircraft Manufacturers Association, American Society for Testing and Materials (ASTM) International, and countless individuals who promoted the concept since the early 1990s. In 2004, the FAA released a rule that created a light-sport classification for airplane, gyroplane, lighter-than-air, weight-shift-control, glider, and powered parachute. [Figure 17-1]



Figure 17-1. The LSA category covers a wide variety of aircraft including: A) airplane, B) gyroplane, C) lighter-than-air, D) weight-shift-control, E) glider, and F) powered parachute.